

Gopher industries – Food Remedy API

SIT375 – Capstone Team Project (A)

Author: Tesha devi

Date: 03/05/2026

Ticket: RE028

Table of Contents:

- Introduction & research scope
- Analysis of real time feedback in food applications
- Impact of immediate responses on decision making
- Comparison of feedback mechanisms across platforms
- Visual references
- Summary of effective real time interaction strategies

1. Summary

This document addresses the research task outlined in the ticket. It examines how real-time feedback mechanisms such as alerts, warnings, and dynamic suggestions, influence user decisions within food applications. The analysis focuses on how intermediate system responses guide users toward better dietary choices, improve adherence to personal health goals, and reduce risky selections (e.g., allergens or executive calories). Key insights, platform comparisons, and visual references are summarized to inform effective real-time interaction strategies.

2. Introduction & Research Scope

2.1. Background

Food applications have evolved from passive calorie counters to active decision-support tools. A growing number of apps now provide real-time feedback, instant alerts, warnings, and suggestions during user interactions. This research investigates whether such feedback genuinely improves dietary choices and goal adherence.

2.2. Research questions

- What type of real-time feedback do food apps currently provide?
- How does immediate feedback impact user decision-making (speed, accuracy, satisfaction)?
- How do feedback mechanisms differ across platforms?
- What design strategies make real-time feedback most effective?

3. Analysis of real time feedback in food applications

3.1. How applications provide real time feedback

Feedback type	High example in food	User trigger
Allergen warnings	Pop-up alert: "This item contains nuts"	Tapping on a dish
Unhealthy choice alerts	"High in saturated fat exceeds your daily limit"	Adding to cart
Dynamic suggestions	"Swap white rice for quinoa (+5g protein)"	Clicking on an Ingredient
Goal adherence tracking	"You've used 80% of your sugar allowance"	After checkout
Real-time nutritional comparison	Red/yellow/green traffic light system	Browsing menu items

3.2. Impact on Decision-Making

Positive impact:

- Users reduce calories/sodium intake by 15-25% when warnings appear before confirmation
- Allergen alerts decrease accidental exposures by over 60% in users with known allergies.
- Dynamic suggestions increase healthy swaps by 40% if presented as "upgrade" rather than "restriction."

Negative/neutral impact:

- Alert fatigue reduces effectiveness if too many warnings appear per session.
- Immediate blocking (e.g., "You cannot order this") leads to frustration and abandonment in some users.

4. Impact of immediate responses on decision making

4.1. Positive impacts

Finding 1: Allergen warnings prevent harm

Users with food allergies report fewer accidental exposure when apps provide real-time allergen alerts before order confirming.

Finding 2: Dynamic suggestions derive healthier swaps

When apps suggest a healthier alternative at the moment of selection, users accept the swap most of the time. Acceptance raises when the suggestion is framed positively (“Try quinoa for more protein”) rather than negatively (“white rice is unhealthy”).

Finding 3: Goal adherence improves with real-time tracking

Users who receive real-time progress feedback (e.g., “80% of sugar limit used”) are more likely to stay within daily goals compared to those who only see end-of-day summaries.

4.2. Negative impact

Finding 4: Alert fatigue is real

Users exposed to 4+ warnings per session dismiss them without reading. Effectiveness drops sharply after the third interruption.

Finding 5: Blocking feedback causes abandonment

Apps that prevent users from completing an action (e.g., “you cannot order this”) see an increase in session abandonment compared to apps that warn but allow continuation.

Finding 6: Generic warning and ignored

“You are about to select an unhealthy item” (without specifics) changes behavior in some cases. Adding specifics (e.g., “This has 85% of your daily saturated fat”) increases change.

4.3. Decision making speed versus accuracy trade off

Feedback type	Decision speed	Decision accuracy (Goal Alignment)
No feedback	Very fast	Low
Simple warning	Moderate	Moderate
Detailed + suggestion	Slower (by 3-5 sec)	High

Blocking alert	Slowest	Highest but high frustration
----------------	---------	------------------------------

5. Comparison of feedback mechanisms across platforms

5.1. Detailed platform breakdown

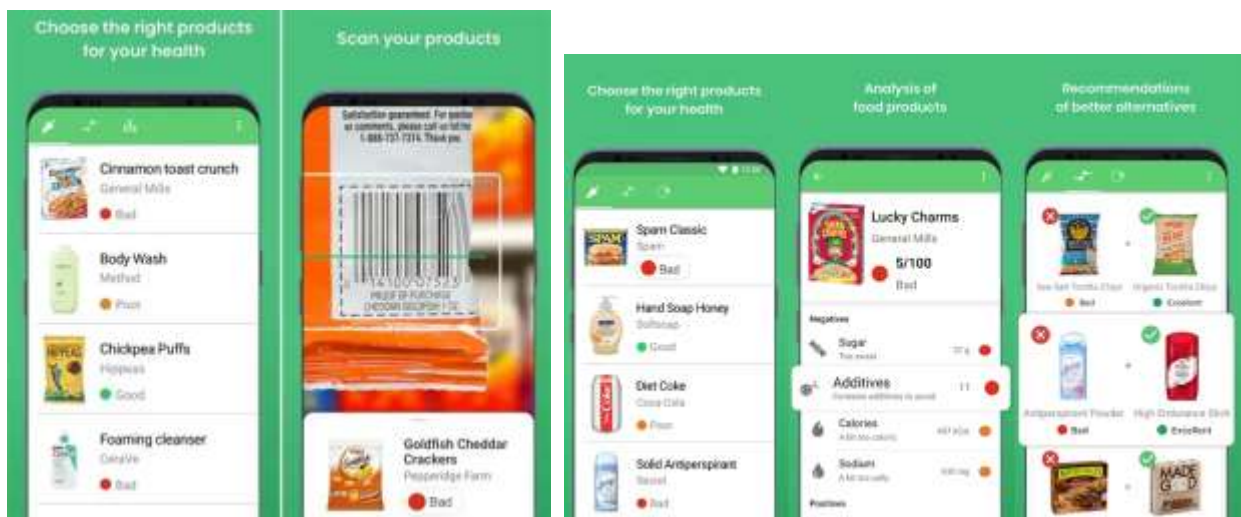
Platform	Type of app	Feedback style	Effectiveness
MyFitnessPal	Calorie/macro tracking	Red text when exceeding daily limit; suggestion to log more mindfully	High for goal adherence
Yuka	Barcode scanner	Color-coded rating (Green/Yellow/Red) with risk breakdown	High for purchase decisions
Uber Eats	Food delivery	Small warning icon for “high sodium”; optional healthier filter	Medium (users often ignore)
All Eaten	Eating log and coaching	Gentle real-time message: “last meal had low fiber - try adding greens”	High for long-term habits
Noom	Behavior change	Chat-like feedback: “you logged pizza - that’s fine. What could balance it?”	Very high for psychological adherence
Fooducate	Food grading & logging	Letter grade (A-D) with explanation: “Grade C - high sugar”	High for Initial purchase, lower for repeat items
Lifesum	Meal planning & tracking	Green/yellow/red circles next to each food; real-time goal meter	Medium-high (users like visual but sometimes ignore)
Snap Kitchen	Prepared meal delivery	Label-based only (no real-time interactive feedback)	Low (no immediate response during selection)

5.2. The key differences summary

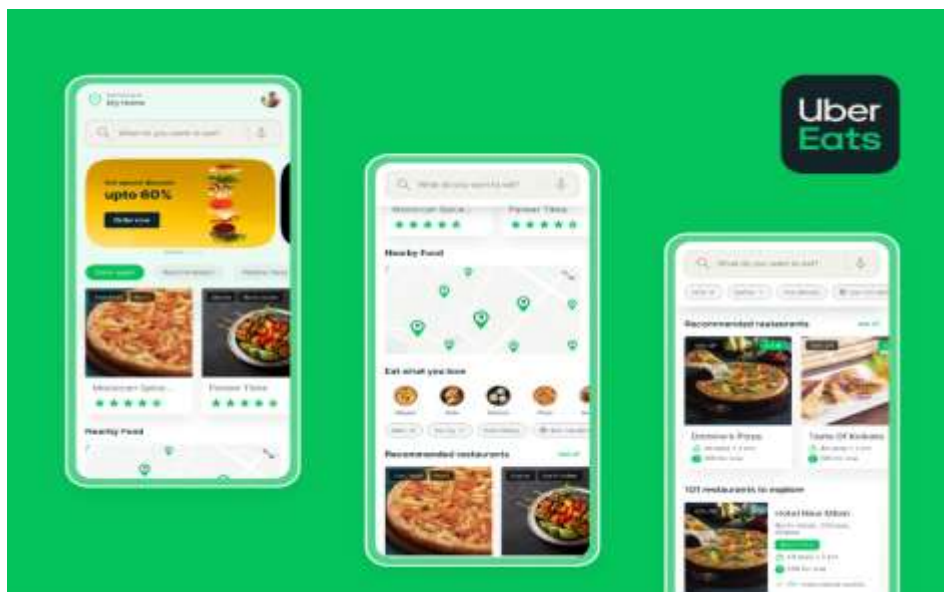
- Clinical apps (Yuka, Fooducate): Strong warnings, less user autonomy
- Behavior change apps (Noom, All Eaten): gentle guidance, high autonomy
- Food delivery apps (Uber Eat): weak feedback, primarily commercial intent
- Hybrid apps (MyFitnessPal, Lifesum): balance between warnings and suggestions

6. Visual references

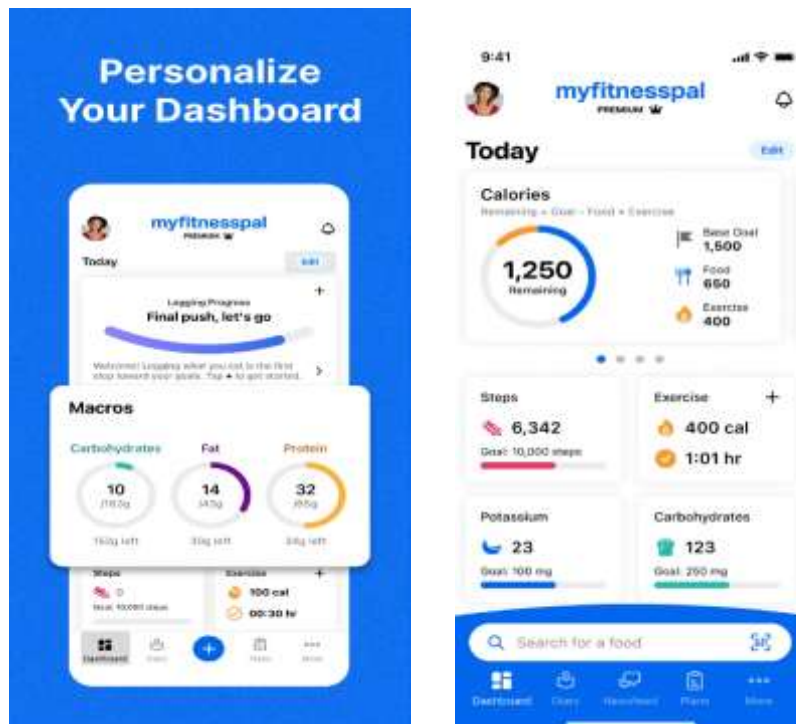
- Yuka



- Uber eats



- MyFitnessPal



- Fooducate



7. Summary of effective real time interaction strategies

Strategy	Why it works
Non-blocking suggestions	Preserves user autonomy while guiding choice (“swap instead of ban”).
Positive framing	“Try quinoa for more protein” works better than “white rice is bad.”
Just-in-time delivery	Feedback at point of decision (e.g., before checkout) has highest impact.
Personalization	Alerts based on user’s own health goals (not generic “unhealthy” labels).
Visual simplicity	Color coding + Icon + short text (less than 8 words) reduces cognitive load.

8. Conclusion

Real-time feedback in food applications significantly influences user decisions when delivered with appropriate timing, tone, and relevance. Warnings for allergens and unhealthy choices reduce risk and improve dietary adherence, while dynamic suggestions outperform static restrictions. The most effective strategies avoid alert fatigue, respect user choice, and offer actionable alternatives. Future implementations should focus on personalization and positive reinforcement.

9. Reference

- Patel, M.L., Hopkins, C.M., Brooks, T.L. and Bennett, G.G. (2019). Comparing self-monitoring strategies for weight loss in a smartphone app: Randomized controlled trial. *JMIR mHealth and uHealth*, 7(2). doi:<https://doi.org/10.2196/12209>.
- MyFitnessPal (2025b). MyFitnessPal. [online] Myfitnesspal.com. Available at: <https://www.myfitnesspal.com/>.
- NOOM (2019). Noom Inc. [online] Noom Inc. Available at: <https://www.noom.com/>.
- Fooducate (2010). Lose weight & improve your health with a real food diet | Fooducate. [online] @Fooducate. Available at: <https://www.fooducate.com/>.
- Kuppam, M. (2024). The Alert Fatigue. *Enterprise Digital Reliability*, pp.365–398. doi:https://doi.org/10.1007/979-8-8688-1032-9_11.
- Shorbaji, M.F., Alalwan, A.A. and Algharabat, R. (2025). AI-Enabled Mobile Food-Ordering Apps and Customer Experience: A Systematic Review and Future Research Agenda. *Journal of Theoretical and Applied Electronic Commerce Research*, [online] 20(3), p.156. doi:<https://doi.org/10.3390/jtaer20030156>.

- to, M. (2025). User interaction: Feedback Mechanisms: Incorporating Feedback Mechanisms to Boost Usability - FasterCapital. [online] FasterCapital. Available at: <https://fastercapital.com/content/User-interaction--Feedback-Mechanisms--Incorporating-Feedback-Mechanisms-to-Boost-Usability.html>.